DevOps

Server Overview

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What is Server?

Server - is a software or hardware device that accepts and responds to requests made over a network.

- The device that makes the request, and receives a response from the server, is called a **client**.
- On the Internet, the term ***"server"*** commonly refers to the computer system that receives requests for a web files and sends those files to the client.
- Any computer, even a home desktop or laptop computer, can act as a server with the right software. For example, you could install an **FTP** server program on your computer to share files between other users on your network.

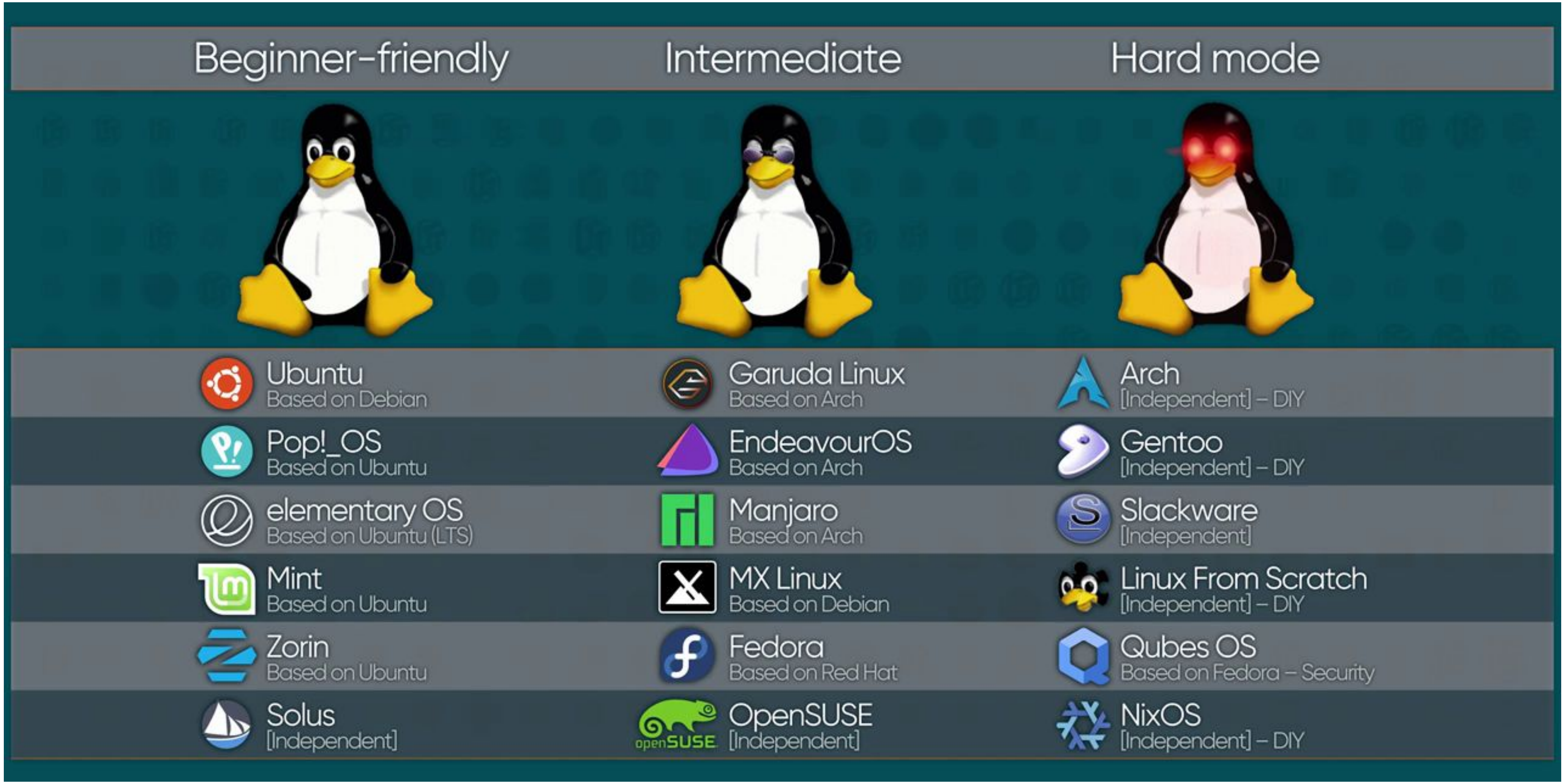
Server Type

- Application server:
- Blade server
- Cloud server
- Database server
- Dedicated server
- Domain name service
- Mail server
- Print server
- Proxy server
- Standalone server
- Web server
- File server

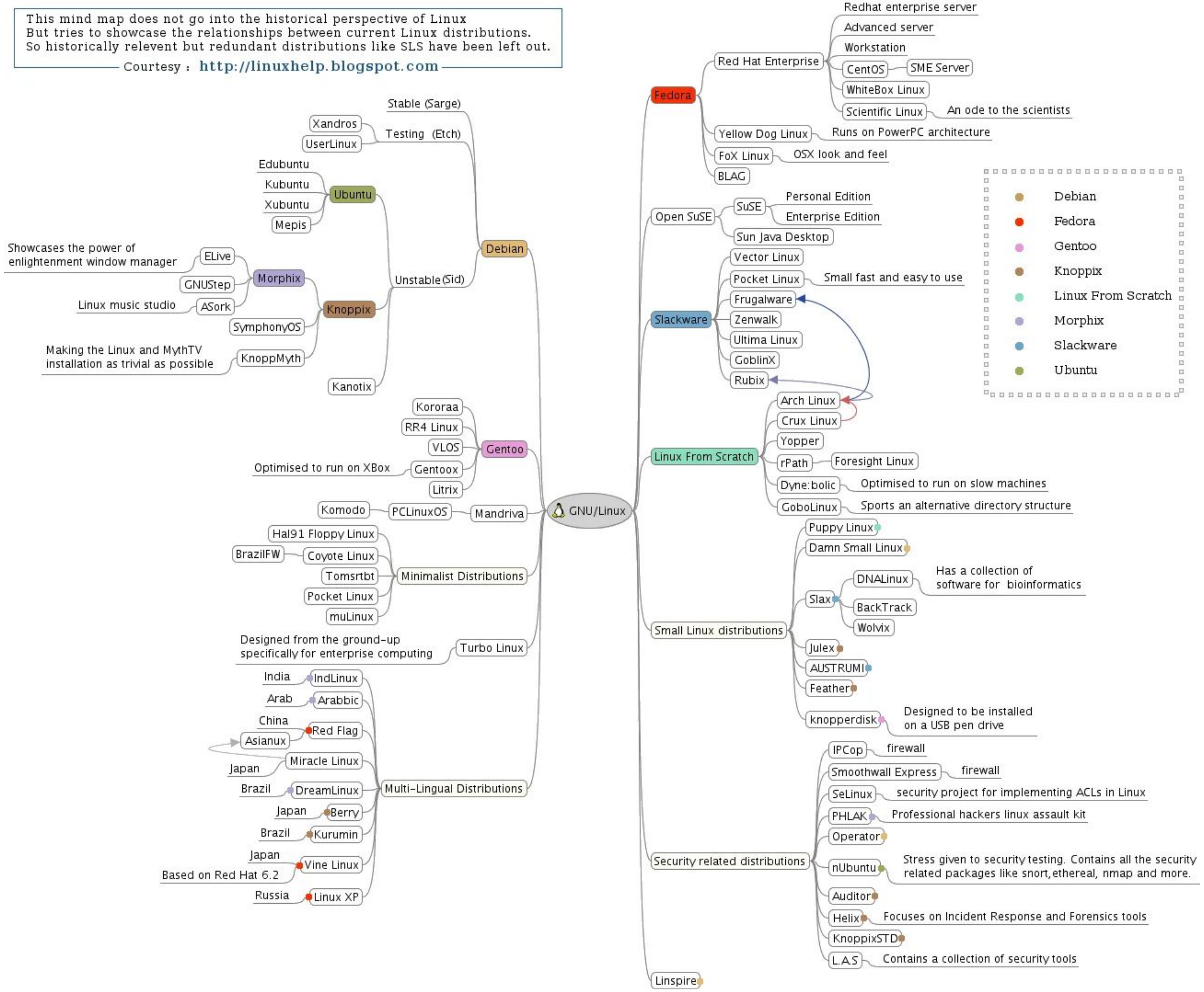
What is Linux?

Linux is a free and open-source operating system that is widely used for a variety of purposes, including servers, desktops, smartphones, and embedded systems. It is based on the Unix operating system and is known for its stability, security, and flexibility. Linux is used by many organizations and individuals due to its open-source nature, which allows anyone to modify and distribute the source code. Linux also provides users with a wide range of applications, including web servers, databases, office productivity tools, and more. Linux is often considered as an alternative to proprietary operating systems such as Microsoft Windows and macOS.

Linux Distribution



Linux Relationship



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Linux Package Manager

A **Linux package manager** is a software tool used to manage the installation, removal, and update of software packages on a Linux system. It provides a centralized system for managing software packages, which makes it easier to install, update, and remove software as needed.

Linux Package Manager

A Linux Some of the most popular Linux package managers include:

- **dpkg** (Debian Package Manager) - Debian Linux family
- **apt** (Advanced Package Tool) – Interface for **dpkg**, Debian Linux family
- **rpm** (RedHat Package Manager) - Redhat Linux family
- **yum** (Yellowdog Updater, Modified) - Interface for **rpm** , Redhat Linux family
- **dnf** (Dandified Yum)- RPM-based distributions, next Generation of YUM.
- **zypper** - package manager on OpenSUSE Linux.
- **snap** - can be used on common major Linux distributions, including Ubuntu, Linux Mint, Debian and Fedora.

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Linux Shell

The Linux shell is a command-line interface for interacting with the operating system. It provides a way for users to enter commands, run scripts, and perform various other tasks on a Linux system.

There are several different shells available for Linux, including:

- **BASH** (Bourne Again SHell): The most widely used shell on Linux, BASH is a powerful and flexible shell that is compatible with the original Bourne shell (SH) and includes many advanced features, such as command history and line editing.
- **ZSH** (Z Shell): A shell that is similar to BASH, but with additional features, such as improved command completion and a more user-friendly interface.
- **CSH** (C Shell): A shell that is similar to the C programming language and provides a syntax that is more familiar to C programmers.
- **TCSH** (Tenex C Shell): An enhanced version of CSH that includes additional features, such as command history and line editing.

Linux Command (files)

- Listing all users : # ***sudo cat /etc/passwd***
- Checking the current working directory : # ***pwd***
- Listing all files and directories as well as hidden files : # ***ls -a***
- Creating new directory : # ***sudo mkdir [dir_name]***
- Going to the desired directory : # ***cd [/path/to/directory]***
- Removing a directory : # ***sudo rm -rf [dir_name][file_name] # rmdir***
- Viewing the content of a file : # ***cat [file_name]***
- Writing a new file : # ***sudo touch [file_name]***

Linux Command (/etc/passwd)

```
james:x:1000:1000::/home/james:/usr/bin/zsh
```

- **james:** This is the username of the account
- **x**: indicates that the password is stored in the **/etc/shadow** for the security reason (stored the user encrypted password)
- **1000**: This is the user ID (UID), it's unique numerical identifier assigned to each user account
- **1000**: This is the group ID (GID), it's a unique numerical identifier assign to each group account
- Value after the 1000, is used to store additional information about the user, such as fullname ...
- **/home/james**: user's home directory
- **/usr/bin/zsh:** This is the absolute path to the user's default shell

Linux Command (/etc/shadow)

```
james:$y$j9T$upItawerweE6OheDX9Bwhr.$zABAnasdfas73B6Ow1IO7asrweqvvJQ39:19758:0:99999:7:::
```

- **Username:** james
- **Encrypted Password**: yj9T$upItawerweE6OheDX9Bwhr.$zABA……
- **Last Password Change**: 19758
- **Minimum Password Age**: 0
- **Maximum Password Age**: 99999
- **Password Warning Period**: 7
- **Password Inactive Period**: 99999
- **Account Expiration Date**: : (empty, meaning the account does not expire)
- **Reserved Field**: : (empty)

Linux Command (file , directory)

- Removing a file : # ***sudo rm -f [file_name]***
- Renaming a file : # ***sudo mv [file_name] [new_file_name]***
- Moving a file from one place to another place :

  # ***sudo mv [/path/to/file/file_name] [/path/to/destination]***
- Copying a file from one place to another place :

  # ***sudo cp -f [/path/to/file/file_name] [/path/to/destination]***
- Copying **all contents** in a directory from source to destination :

  # ***sudo cp -R [/path/to/dir/dir_name/] [/path/to/destination]***

Linux Command (user ,group)

- Adding new user : # ***sudo useradd [username]***
- Changing user's password : # ***sudo passwd [username]***
- Logout from current user : # ***exit*** | ***logout***
- Login or switching to a user account : # ***sudo su [username]***
- Adding user to a group : # ***sudo useradd [username] [groupname]***
- Or # ***sudo usermod -aG [groupname] [username]***
- Deleting a user from group : # ***sudo deluser [username] [groupname]***
- Visiting a website : # ***curl [website_address]***
- Editing an existing file : # ***sudo vim [file_name]***
- Checking disks usage in system : # ***sudo df -h***

Linux Command

- Updating server : # ***`sudo apt update`***
- Upgrading server : # ***`sudo apt upgrade`***
- Installing tool or software : # ***`sudo apt install -y [package_name]`***
- Starting service : # ***`sudo systemctl start [service_name]`***
- Checking service status : # ***`sudo systemctl status [service_name]`***
- Stopping service : # ***`sudo systemctl stop [service_name]`***
- Auto start service: # ***`sudo systemctl enable [service_name]`***
- Disable auto start service: # ***`sudo systemctl disable [service_name]`***
- Reloading a service : # ***`sudo systemctl reload [service_name]`***
- Reloading the server(Refresh) : # ***`sudo systemctl daemon-reload`***

Thank you

Begin with the theory, Start with the practical